SEMESTER 5 – WEEK 5 – Graphs and Data Interpretation

Why Scientific Decisions Depend on Data Literacy

Over the last twenty years, the scientific community has entered what many researchers call *the age of data dependence*. With the expansion of digital sensors, satellite imaging, laboratory automation, and online databases, scientists are no longer limited by a lack of information. Instead, they are challenged by the difficulty of analysing enormous and complex datasets. In fields such as environmental science, epidemiology, and urban planning, data literacy—not simply reading numbers, but interpreting their meaning – has become a core scientific competence.

For science students, developing strong data-interpretation skills is essential. A graph is not just a visual decoration; it is a simplified model of reality. Students must learn to examine several features before accepting the message a graph appears to show. First, they should **read the axes carefully**, paying attention to units, time intervals, and measurement scales. Even small changes in scale can dramatically alter the appearance of a trend. Second, students must **identify the variables** represented and determine whether they are continuous (like temperature) or categorical (like types of pollutants). This affects which type of graph is appropriate.

Science students should also be able to **distinguish correlation from causation**. Two lines that rise together do not necessarily indicate that one variable causes the other to change. Careful readers must look for missing variables, unrecorded factors, or gaps in the data. In addition, students should learn to question **how the data was collected**. Was it gathered evenly across regions? Was the sample size sufficient? Were instruments calibrated properly? Understanding these aspects helps prevent misinterpretation.

Graphs and charts simplify complex information, but they also shape the way readers perceive it. A line graph showing annual pollution levels in Morocco's industrial zones might suggest a steady increase, but a closer look at the vertical scale may reveal that the change is relatively small. Likewise, a bar chart comparing population densities across Moroccan regions highlights differences, but these differences depend heavily on measurement units and data-collection methods. As a result, scientists must not only interpret visual patterns but also question the reliability and limitations of the data.

Another challenge is the growing use of predictive models. These models rely on large datasets to estimate future trends—such as coastal erosion, rainfall variability, or population expansion. While such models are powerful, they can be misleading if the underlying data is incomplete or biased. For instance, rural areas may lack consistent data collection, causing models to emphasise urban trends. This can lead policymakers to make decisions that favour well-documented regions while neglecting less-studied ones.

Authentic data literacy therefore requires a combination of skills: understanding how data is collected, knowing how to visualise it effectively, interpreting patterns critically, and recognising uncertainty. Science students must approach graphs as analytical tools rather than simple illustrations. Readers should ask: *What does this graph show? What does it hide? What assumptions does it make? What alternative explanations exist?* Such reflective questioning is central to scientific thinking.

In Morocco, as in many countries, strengthening data literacy is vital for informed decision-making. Whether the goal is to confront environmental pressures, manage water resources, or plan sustainable urban growth, policymakers and scientists depend on accurate interpretations of visual data. For science students, this means approaching every graph with a critical and questioning mindset – recognising what the data reveals, what it might conceal, and how confidently conclusions can be drawn. Developing these habits early prepares students to contribute responsibly to evidence-based scientific work and national development.

I. Comprehension

1. Multiple Choice – Global Understanding

Choose the best answer:

The main purpose of the text is to show that data literacy is:

a) Only necessary for mathematicians.
b) Mainly about learning computer programming.
c) A fundamental skill for interpreting scientific information.
d) Less important than collecting large amounts of data.

2. Short Answer – Key Details

Answer in one or two sentences.

1. What two features of a graph must students check before accepting its message?
2. How can gaps in data influence scientific conclusions?

3. Table Completion – Types of Problems Mentioned

Fill in the table with information from the text.

Problem Type	Example from Text
Graph Design Issue	
Data Collection Issue	
Interpretation Issue	

4. Sentence Completion – Understanding Cause/Effect

Complete the sentences using information from the text.

1. Predictive models may become misleading when _____.
2. A trend on a graph might look stronger than it really is because _____.

5. True/False/Not Mentioned

1. All Moroccan regions collect scientific data with equal regularity.
2. A line graph may exaggerate or minimise a trend depending on its scale.
3. The text states that correlation always indicates causation.
4. Students should examine how data was collected before trusting a graph.

6. Inference – Reading Between the Lines

1. What does the author suggest about the reliability of visual patterns alone?
2. Why might science students need to question what a graph *hides*?

7. Critical Reflection

Imagine you are a policymaker using limited data. What risks could arise if you base decisions only on urban datasets?

8. Application Task – Real-World Problem Solving

You receive a bar chart showing a sudden rise in air-pollution levels. List **three** checks you should make before accepting the result.

II. Vocabulary Development

1. Match the terms with their definitions:

1) categorical variable _____
2) correlation _____
3) dataset _____
4) measurement scale _____
5) calibration _____
6) predictive model _____

a) A collection of related data points, usually organised in tables, files, or databases so they can be analysed.
Example: A spreadsheet containing the monthly rainfall for each Moroccan city is a dataset.
b) The process of adjusting or fine-tuning an instrument or measurement tool so that it gives correct and reliable results. *Example:* A scale must be calibrated before weighing chemicals in a laboratory.
c) A type of variable that represents groups, categories, or labels rather than numerical values.
Example: "Type of smartphone" (Samsung, Apple, Huawei) is a categorical variable.
d) A statistical relationship between two variables that shows how they change together – either increasing, decreasing, or showing no clear pattern.
Example: There is a strong correlation between hours studied and exam scores.
e) A mathematical or statistical system used to estimate or forecast future outcomes based on existing data.
Example: Weather apps use predictive models to forecast tomorrow's temperature.
f) A defined range, unit, and spacing used to measure and interpret numerical data on instruments or graphs.
Example: A thermometer with markings every 1°C uses a specific measurement scale.

2. Word-Formation Practice:

Use the following words to form a noun, adjective, or verb that completes each sentence.

interpret – dependent – reliable – visual – analyse – collection

1. Strong data _____ is essential for scientific work.
2. The graph provides a clear _____ summary of the results.
3. Scientists must question the _____ of data sources.
4. Moroccan researchers are increasingly _____ on satellite data.
5. Students learn to _____ graphs more accurately with practice.
6. Poor data _____ methods can affect scientific conclusions.

III. Grammar – Using Comparative Structures in Scientific Writing

A. Rewrite the sentences using the comparative form:

1. Urban pollution levels are high, but industrial-zone pollution levels are even higher.
   → Industrial-zone pollution levels are _____ urban pollution levels.
2. The 2024 model is efficient, but the 2025 model is efficient too.
   → The 2025 model is _____ the 2024 model.
3. Rural data coverage is limited, and coastal data coverage is extensive.
   → Coastal data coverage is _____ rural data coverage.

B. Complete each statement with an appropriate comparative expression (more / less./ fewer + noun):

1. Regions with _____ monitoring stations often produce incomplete datasets.
2. A graph becomes clearer when it has _____ unnecessary labels.
3. Students make _____ mistakes when they read axes carefully.

IV. Speaking – Group Debate: "Can Data Ever Be Neutral?"

Form groups of 3 – 4 students. Read the discussion prompt below, then debate the questions. Each student should take a position, defend it with examples, and respond to others' arguments.

Scientists often treat data as objective, but every dataset is influenced by choices – what to measure, who to include, which variables to ignore, and how the results are visualised. This raises an important question: *Is any data truly neutral, or is it always shaped by human decisions?*

Debate Questions:

1. To what extent can data be considered objective?
2. Who decides what counts as "important data"?
3. Can a graph or chart be accurate but still misleading?
4. Is it ever acceptable to simplify data for the public?
5. What responsibilities do scientists have when presenting results to society?
6. Can two scientists interpret the same dataset differently and both be right?
7. How can students develop "healthy skepticism" when reading scientific graphs?

Groups summarise their main points and present a **short collective statement** addressing the central question: **"Can Data Ever Be Neutral?"**

V. Writing – Analytical Paragraph: "Describe and Interpret a Scientific Graph"

Choose a graph and write a **7 – 10 sentence analytical paragraph**. Your paragraph must:

- Identify the type of graph and the variables shown.
- Describe the main trend(s) accurately.
- Include **at least two comparative structures**.
- Comment on data reliability, scale, or missing information.
- End with a cautious statement about the confidence level of your interpretation.

Use clear scientific language and avoid simply listing numbers. Focus on interpretation, not transcription.